

Part A: JavaScript Logic

Instructions: Write code and test in Console.

Task 1: The Voting Machine (Comparison)

Create a variable age. Write an if/else condition:

- If age is **18 or more**, print "You can vote".
- If age is **less than 18**, print "You are too young".
- **Hint:** "Check karne ke liye hum Greater Than or Equal operator ($\geq$) use karte hain. if (age $\geq$ 18) try karein."

Task 2: Area Calculator (Math Function)

Write a function calculateArea(width, height) that multiplies two numbers and **returns** the result.

- Call the function with 10 and 5, and print the answer.
- **Hint:** "Function ke andar console.log nahi, balkay return width * height likhein. Phir function call karte waqt result ko aik variable mein save karein."

Task 3: Even Number Printer (Loop with Logic)

Write a Loop that prints only **Even Numbers** from **1 to 10** (e.g., 2, 4, 6...).

- **Hint:** "Loop ko normal chalayen. Loop ke andar aik if lagayen. Agar $i \% 2 === 0$ hai, to sirf tab hi console.log(i) karein."

Task 4: The Array Looper (Iteration)

Create an array: ["Apple", "Banana", "Cherry"].

Write a Loop that prints each fruit **one by one** in the console.

- **Hint:** "Loop ko 0 se start karein aur array.length tak chalayen. Console mein fruits[i] print karein taaky har dafa agla item aye."

Task 5: Profile Update (Adding Properties)

Create an object: user = { name: "Ali" }.

Add a **new property** called city with value "**Karachi**" to this object (without changing the original line).

- **Hint:** "Nayi property add karna waisa hi hai jaise purani update karna. Bas user.city = 'Karachi' likh kar dekhein."

Task 6: The Sum Machine (Accumulator)

Create an array of numbers: [10, 20, 30].

Write a loop to calculate the **Total Sum** (60) and print it.

- **Hint:** "Loop se pehly aik variable let total = 0 banayen. Loop ke andar total = total + array[i] likhein taaky har number purani value mein jama hota jaye."
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Part B: DOM Manipulation

Task 7: The Square Calculator

Create an Input field (type number) and a Button "Calculate Square".

When clicked, take the number, multiply it by itself (e.g., 5 * 5), and show the result in an alert().

- **Hint:** "Input se value hamesha string milti hai. Calculation ke liye input.value * input.value karein, JS khud hi math kar lega."

Task 8: The Magic Toggler (Hide/Show)

Create a Box (div) and a Button "Toggle".

- If the box is visible, clicking the button should **hide** it.
- If the box is hidden, clicking the button should **show** it.
- **Hint:** "if/else lagayen. Check karein ke agar div.style.display === 'none' hai to usay 'block' kar dein, warna 'none' kar dein."

Task 9: Valid Input Checker (Validation)

Create an Input field and a Button "Submit".

- If the input is **empty**, change the input's border color to **Red**.
- If the input has text, change the border color to **Green**.
- **Hint:** "Input ki value check karein: if (input.value === ''). Border change karne ke liye style.borderColor use hota hai."

Task 10: Image Swapper (Bulb On/Off)

Create an img showing a **Black Bulb** (Off). Create two buttons: "On" and "Off".

- "On" button changes image to a **Yellow Bulb**.
- "Off" button changes image back to **Black Bulb**.
- **Hint:** "Apko bas Image ka src attribute change karna hai. img.src = 'yellow-bulb.jpg' try karein."

Task 11: The List Maker (Create Element)

Create a Button "Add Item". When clicked, add a new with text "New Task" inside an existing .

- **Hint:** "Ye 3 steps yaad rakhein: 1. document.createElement('li'), 2. li.innerText = 'New Task', 3. ul.appendChild(li)."

Task 12: The Click Counter (State)

Create an <h1> that shows the number **0**. Create a Button "Count".

Every time the button is clicked, the number should increase (0 -> 1 -> 2...).

- **Hint:** "Function ke bahar aik variable let count = 0 banayen. Jab button click ho, count++ karein aur phir H1 ka innerText update karein."